

Homework 22: Properties of Determinants

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BCS A, E, F, G

Theorem 1. *Determinant of a Matrix Product.*

If A and B are square matrices of order n , then $\det(AB) = \det(A) \det(B)$.

Theorem 2. *Determinant of a Scalar Multiple of a Matrix.*

If A is a square matrix of order n and c is a scalar, then the determinant of cA is

$$\det(cA) = c^n \det(A).$$

Theorem 3. *Determinant of an Invertible Matrix.*

A square matrix A is invertible (nonsingular) if and only if $\det(A) \neq 0$.

Theorem 4. *Determinant of an Inverse Matrix.*

If A is an $n \times n$ invertible matrix, then $\det(A^{-1}) = 1/\det(A)$.

Theorem 5. *Determinant of a Transpose.*

If A is a square matrix, then $\det(A) = \det(A^T)$.

Equivalent Conditions for a Nonsingular Matrix. If A is an $n \times n$ matrix, then the statements below are equivalent.

1. A is invertible.
2. A is row-equivalent to I_n .
3. A can be written as the product of elementary matrices.

Problem 1. *The Determinant of a Matrix Product.*

Find $|A|$, $|B|$, and $|AB|$ for the matrices

$$A = \begin{bmatrix} 1 & -2 & 2 \\ 0 & 3 & 2 \\ 1 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 & 1 \\ 0 & -1 & -2 \\ 3 & 1 & -2 \end{bmatrix}.$$

Problem 2. *The Determinant of a Scalar Multiple of a Matrix.*

Find the determinant of the matrix.

$$A = \begin{bmatrix} 10 & -20 & 40 \\ 30 & 0 & 50 \\ 20 & -30 & 10 \end{bmatrix}.$$

Problem 3. *Classifying Square Matrices as Singular or Nonsingular.*

Determine whether each matrix has an inverse.

$$A = \begin{bmatrix} 0 & 2 & -1 \\ 3 & -2 & 1 \\ 3 & 2 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 2 & -1 \\ 3 & -2 & 1 \\ 3 & 2 & 1 \end{bmatrix}.$$

Problem 4. The Determinant of the Inverse of a Matrix.

Find $|A^{-1}|$ for the matrix A .

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & -1 & 2 \\ 2 & 1 & 0 \end{bmatrix}.$$

Problem 5. The Determinant of a Transpose.

Show that $|A| = |A^T|$ for the matrix below.

$$A = \begin{bmatrix} 3 & 1 & -2 \\ 2 & 0 & 0 \\ -4 & -1 & 5 \end{bmatrix}.$$

Problem 6. Systems of Linear Equations.

Which of the systems has a unique solution?

a. $2x_2 - x_3 = -1$, $3x_1 - 2x_2 + x_3 = 4$, $3x_1 + 2x_2 - x_3 = -4$.

b. $2x_2 - x_3 = -1$, $3x_1 - 2x_2 + x_3 = 4$, $3x_1 + 2x_2 + x_3 = -4$.

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Good Luck